Informatics II, Spring 2026, Solution 0

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Task 2

The algorithm searches the array, and -3 is at position 3 in the array

Task 3

Algo: Find_Second_Largest(A)

$largest = MIN$;
$second\_largest = MIN$;
for *each* $elem$ *in* *A* do
  if $elem > largest$ **then** $largest := elem$;
for *each* $elem$ *in* *A* do
  if $elem > second\_largest$ ***and*** $elem \neq largest$ **then** $second\_largest := elem$;

```c
#include <stdio.h>

int main() {
  int n = 5;
  int A[] = {11, 3, -3, 2, -5};

  int largest = -99999;
  for(int i = 0; i < n; i++) {
    if(A[i] > largest) {
      largest = A[i];
    }
  }

  int sec_largest = -99999;
  for(int i = 0; i < n; i++) {
    if(A[i] > sec_largest && A[i] != largest) {
      sec_largest = A[i];
    }
  }
  printf("%d\n", sec_largest);
}
```

Task 4

```c
#include <stdio.h>

int main() {
    int A[4][4] = {
        {2, 4, 1, 3},
        {5, 7, 8, 2},
        {9, 6, 4, 1},
        {3, 8, 5, 6}
    };
    int sum = 0;
    for (int i = 0; i < 4; i++){
        for (int j = 0; j < 4; j++){
            if(i==j)sum += A[i][j];
        }
    }
    printf("%d\n", sum);
}
```